Can He

hecanhappy@gmail.com | (617) 513-2316 | Boston, MA | bostonsuprememantou.github.io

PROFESSIONAL EXPERIENCE

Tristar AI Boston, MA

Computer Vision Engineer May 2026 - Present

- Own development of new vision pipelines and maintenance of existing ones, embedding automation services to minimize manual intervention. Fully automated vision workflows across multiple production lines, including model training and deployment.
- Partner with Customer Success to turn new inspection requirements into working vision demos within hours, owning solution design and implementation for early technical evaluation.

Software Development Intern Jun 2025 - Jan 2026

- Owned end-to-end defect detection for four categories in a six-camera, three-line production system, covering preprocessing, inference integration, postprocessing, line-specific decisions, and PLC rejection.
- Designed a manager-worker runtime with configurable process allocation to control memory use and avoid redundant workers; added pytest and failure-path tests across camera, inference, messaging, and storage boundaries.

Northeastern University Boston, MA

Research Assistant Dec 2024 - Jul 2025

- Built preprocessing and data-quality workflows for multi-state road-network, weather, and accident datasets under Professor Hongyang Zhang, reconciling heterogeneous records and resolving missing or inconsistent fields for traffic safety research.

Teaching Assistant, DS5220 Supervised Machine Learning Sep 2024 - Jun 2025

- Supported DS5220 across two semesters through office hours, review sessions, grading, and answer-key preparation; explained supervised learning concepts and provided feedback.

Wuhu Automobile Advanced Technology Institute Co., Ltd. Wuhu, China

Project Manager Sep 2020 - Sep 2023

- Led a 23-person engineering team developing carbon-fiber automotive components, reducing targeted component weight by 42%.
- Coordinated design feasibility, testing, cost control, and production integration to meet manufacturing requirements.

Chery Holding Group Co., Ltd. Wuhu, China

Project Manager Sep 2016 - Sep 2020

- Managed chassis adaptation and testing for gasoline and hybrid vehicles, coordinating model updates and engineering data.
- Partnered with quality and production teams on technical validation, patent-related documentation, and manufacturing guidance.

SELECTED PROJECTS

PaddlePaddle GPU Compatibility and On-Demand Inference Service Personal Project

- Submitted upstream PaddlePaddle PR #79331 for Blackwell sm_120 build configuration, including CUDA-version guards, Thrust C++ compatibility fixes, and regression tests.
- Built an inference service around a personal PaddlePaddle fork with on-demand model loading and GPU-memory release after 60 seconds of inactivity; subsequent requests reload the model.

Agent-Driven Vision Pipeline Development and Release Tristar AI | In Development

- Implemented a prompt-driven workflow that calls a reusable engineering toolbox to assemble complete inspection pipelines and produce releases, with two-stage engineering review before release approval.
- Trained reusable defect-detection models and made them available alongside SAM3 and EfficientAD through agent-callable tools.

Tick-Level Quantitative Research and Workflow Platform Personal Project

- Built **Lyre** for Arrow/Parquet tick/bar storage and SQLite metadata, and **Argus** for time-aligned, no-lookahead features.
- Designed SQLite tables, primary-key and uniqueness constraints, and indexes; implemented parameterized queries and transactional updates with idempotent request handling.
- Built dependency-aware research and backtest orchestration with retries, timeouts, atomic state publication, and schema and freshness checks before downstream execution.
- Built authenticated command APIs with explicit states, expiry, and receipt reconciliation for submission and recovery.

TECHNICAL SKILLS

Languages: Python, SQL, C++
Data & Backend: SQLite, PyArrow, Parquet, Polars, Pandas, HTTP APIs, RabbitMQ, AWS S3
AI & Computer Vision: PyTorch, OpenCV, PaddlePaddle, model integration, inference pipelines
Engineering & Deployment: Docker, Grafana alerting, Git, pytest, multiprocessing, integration testing

EDUCATION

Northeastern University | M.S. in Computer Science | GPA: **3.94/4.00** Apr 2026
Northeastern University | M.S. in Project Management Jan 2016
Chongqing University of Technology | B.E. in Mechanical Engineering Jun 2012